Authentication Log Analysis

Practical Report — Cybersecurity using Microsoft Excel Online

1. Aim

To analyze synthetic authentication logs using Microsoft Excel Online, count successful and failed login attempts, and identify IP addresses with repeated failed attempts that may require further investigation.

2. Tools Used

Microsoft Excel Online, a web browser, and a synthetic CSV dataset. No Python installation was required for this Excel-based practical.

3. Dataset Description

The dataset contains 20 sample login events with four fields: timestamp (date/time), username (account name), source_ip (origin of the attempt), and status (Success or Failed). The records are synthetic and intended for learning only.

4. Procedure

1. Opened the CSV dataset in Excel Online.

2. Reviewed the timestamp, username, source IP, and login status columns.

3. Used COUNTIF to count successful logins: =COUNTIF(D2:D21,"Success").

4. Used COUNTIF to count failed logins: =COUNTIF(D2:D21,"Failed").

5. Used COUNTIF and COUNTIFS to count occurrences and failed attempts for IP address 203.0.113.50.

6. Reviewed the results and flagged repeated failed attempts for investigation.

5. Results

Metric	Result
Total login events	20
Successful logins	7
Failed logins	13
Occurrences of IP 203.0.113.50	6
Failed attempts from IP 203.0.113.50	6

6. Findings and Interpretation

The IP address 203.0.113.50 produced six failed login attempts, exceeding the practice threshold of five failures. This pattern may indicate repeated password guessing or a user/system issue. It is a warning sign, not proof that an attack occurred; further investigation would be needed.

7. Conclusion

The Excel analysis successfully summarized login outcomes and identified a source IP with repeated failed login attempts. This practical demonstrates how basic spreadsheet formulas can support initial authentication-log monitoring and help highlight activity for review.

8. Interview Questions

Q1. What indicates suspicious authentication activity?
Repeated failed attempts, especially from one IP address or against one account, can be suspicious and should be reviewed.

Q2. Why monitor failed logins?

To help detect password guessing, brute-force attempts, and possible unauthorized access.

9. Practical Screenshot

Screenshot of the completed Excel Online analysis, showing the dataset and summary counts.

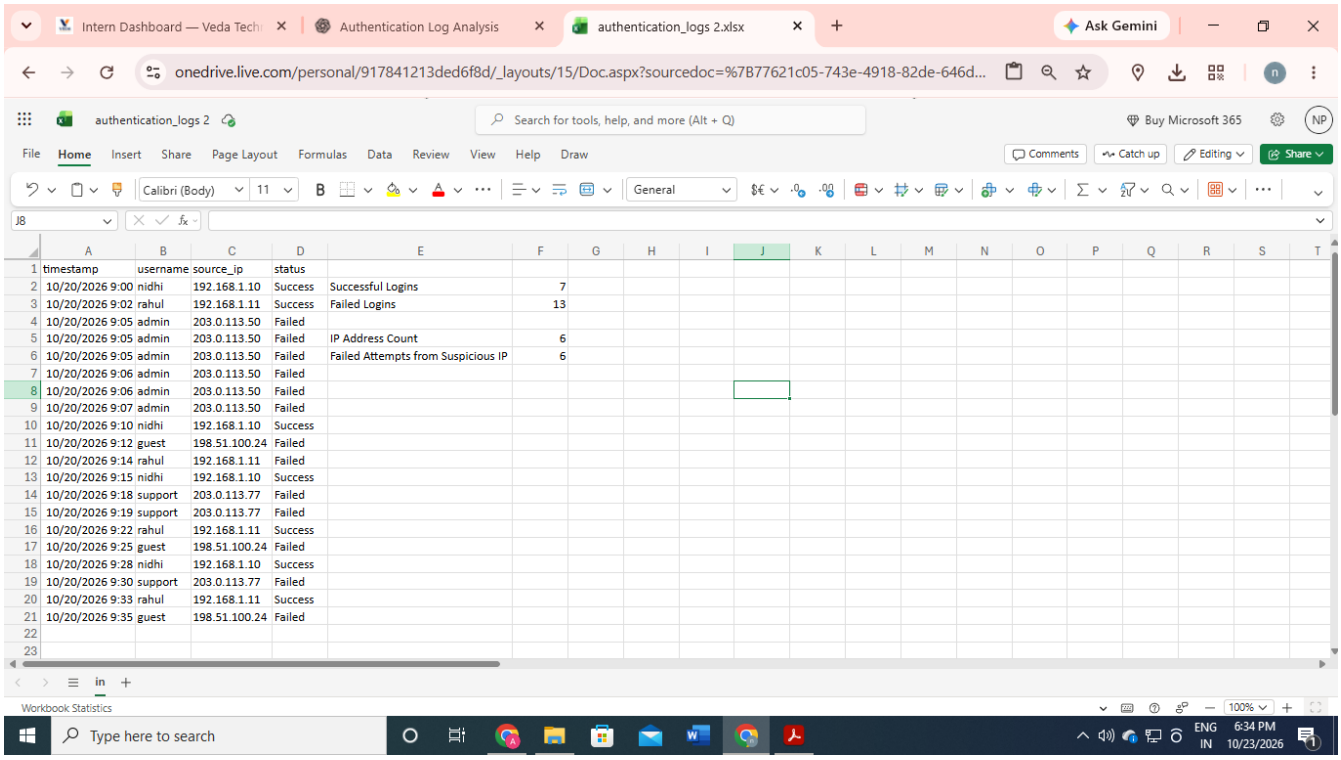


Figure 1: Authentication log dataset and summary counts in Excel Online.